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Zhanibek Kozhirbayev

Date of Birth: 1989
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Profile

Postdoctoral-level researcher in Speech Processing and Natural Language Processing for Low-Resource Languages, with a strong focus on speech recognition, speech translation, and multilingual NLP. I hold a PhD in Computer Science and have 12+ years of combined academic and industrial experience. My research integrates deep learning, self-supervised learning, and cloud/HPC infrastructures, with an emphasis on reproducible research and real-world deployment. I have led and contributed to nationally funded research grants, published in Q1–Q2 journals, and actively collaborate internationally.

Educational

PhD Computer Science 2016 – 2019
Institution: **L.N.Gumilyov Eurasian National University, Astana**
Thesis: "Spoken Term Detection for the Kazakh language on the Cloud"
Visiting Researcher: ACE Lab, University of Texas at San Antonio, USA

Master of Information Technology (Distributed Computing) 2013 – 2015
Institution: **The University of Melbourne, Australia**
Thesis: "Comparison of Microservice Hosting Technologies for the Cloud"

Bachelor of Computing science and software 2006 – 2010
Institution: **L.N.Gumilyov Eurasian National University, Astana**

Work Experience

National Laboratory Astana, Nazarbayev University, Kazakhstan
From 10/15-present
Leading Researcher
Research and development of various tasks of NLP and speech recognition for the Kazakh language:

- Lead research on ASR, speech synthesis, and NLP for the Kazakh language
- Fine-tuning and evaluation of wav2vec2.0, Whisper, mBART, and end-to-end speech translation models
- Development of spoken term detection and low-resource MT systems
- Supervision of junior researchers and graduate students
- Design and maintenance of private research cloud infrastructure (OpenStack)
- Deployment of distributed AI workloads on Hadoop, Kubernetes, DGX-1 clusters

TengriLab, Astana, Kazakhstan
10/2020 – 05/2022
Senior Data Scientist

- Various NLP and Speech Recognition tasks
- Putting DL/ML models into production (computer vision models)

- AI Inference in Production with NVIDIA Triton server
- Technology stack: TensorRT, TritonServer, keras, tensorflow, pytorch, IBM Power9 servers

Astana International University, Astana, Kazakhstan
09/2018 – 12/2018
Senior lecturer
- Taught Programming and Problem Solving

ACE laboratory, the University of Texas at San Antonio, Texas, USA
02/2018 – 05/2018
PhD researcher
- Speaker Recognition for Robotic Control via an IoT Device
- Robotic control
- Neural networks, machine learning

Clarinox Technologies Pty Ltd, Melbourne, Australia
07/14-10/14
Software tester
- Pro-actively participated in weekly team meetings with software developers to maximize productivity and efficiency
- Supported product rollouts across multiple code releases
- Created test cases for a product development
- Continuously updated testing documentation

Skills

Programming	Python, C++
ML/DL	PyTorch, TensorFlow, Keras, Fairseq, ESPnet, NeMo, HuggingFace
Speech & NLP	Kaldi, wav2vec2.0, Whisper, mBART, Tacotron2, VITS
Cloud & HPC	OpenStack, AWS, Kubernetes, Hadoop, DGX systems
Tools	Elasticsearch, Docker, Triton Inference Server

Professional Activities

Projects

- "Development of methods and algorithms for multilingual and multimodal speech-to-speech conversion for academic lectures and talks", scientific grant from the Ministry of Science and Higher Education of the Republic of Kazakhstan for 2024-2026, Role: **principle investigator**.
- "Development of a speech translation system for two low-resource languages through some intermediate language", scientific grant from the Ministry of Education and Science of the Republic of Kazakhstan for 2022-2024, Role: **principle investigator**.
- "Development of unsupervised methods and algorithms for speech recognition based on unaligned audio and text data", scientific grant from the Ministry of Education and Science of the Republic of Kazakhstan for 2020-2022, Role: **senior researcher**.
- "Development of freely distributed open source software tools for automatic text processing in the Kazakh language", scientific grant of the Ministry of Education and Science of the Republic of Kazakhstan for 2018-2020, Role: **researcher**.
- "Development of systems for collecting, processing and analyzing big data (Big Data) using cloud technologies to improve labor efficiency and decision making", scientific and technical program of the Ministry of Education and Science of the Republic of Kazakhstan for 2015-2017. Role: **researcher**.

Selected publications

Journals

- **Kozhirbayev, Z.,** & Yessenbayev, Z. (2025). Fine-tuning methods and dataset structures for multilingual neural machine translation: A Kazakh–English–Russian case study in the IT domain. *Electronics, 14*(15), 3126. DOI: https://doi.org/10.3390/electronics14153126 (WoS: Q2, IF: 2.9; Scopus: citescore: 6.1, percentile: 76)
- **Kozhirbayev, Z.,** & Yessenbayev, Z. (2024). Semantically expanded spoken term detection. *IEEE Access, 12*, 177844–177855. DOI: https://doi.org/10.1109/ACCESS.2024.3506982 (WoS: Q2, IF: 3.4; Scopus: citescore: 9.0, percentile: 86)
- Islamgozhayev, T., Amirgaliyev, B., & **Kozhirbayev, Z.** (2024). Enhancing skeleton-based action recognition with hybrid real and gan-generated datasets. *Eastern-European Journal of Enterprise Technologies, 6*(2), 14–22. DOI: https://doi.org/10.15587/1729-4061.2024.317092 (Scopus: citescore: 2.4, percentile: 47)
- **Kozhirbayev, Z.** (2024). Enhancing neural machine translation with fine-tuned mBART50 pre-trained model: An examination with low-resource translation pairs. *Ingenierie des Systemes d'Information, 29*(3), 831–838. DOI: https://doi.org/10.18280/isi.290304 (Scopus: citescore: 2.4, percentile: 43)
- **Kozhirbayev, Z.** (2023). Kazakh speech recognition: Wav2vec2.0 vs. Whisper. *Journal of Advances in Information Technology, 14*(6), 1382–1389. DOI: https://doi.org/10.12720/jait.14.6.1382-1389 (WoS: Q4, IF: 0.9; Scopus: citescore: 4.8, percentile: 64)
- **Kozhirbayev, Z.,** & Islamgozhayev, T. (2023). Cascade speech translation for the Kazakh language. *Applied Sciences, 13*(15), 8900. DOI: https://doi.org/10.3390/app13158900 (WoS: Q1, IF: 2.5; Scopus: citescore: 5.5, percentile: 69)
- Zulkhazhav, A., **Kozhirbayev, Z.,** Yessenbayev, Z., & Sharipbay, A. (2019). Kazakh text summarization using fuzzy logic. *Computacion y Sistemas, 23*(4), 851–859. DOI: https://doi.org/10.13053/CyS-23-3-3239 (WoS: Q4, IF: 0.4; Scopus: citescore: 1.0, percentile: 14)
- **Kozhirbayev, Z.,** & Sinnott, R. O. (2016). A performance comparison of container-based technologies for the cloud. *Future Generation Computer Systems, 68*, 175–182. DOI: https://doi.org/10.1016/j.future.2016.02.007 (WoS: Q1, IF: 6.1; Scopus: citescore: 17.1, percentile: 97)

International conferences

- Borisov, M., **Kozhirbayev, Z.,** & Malykh, V. (2025). Low-resource machine translation for code-switched Kazakh–Russian language pair. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2025), Student Research Workshop* (Vol. 4, pp. 66–76). Albuquerque, NM, United States. Association for Computational Linguistics. DOI: https://doi.org/10.18653/v1/2025.naacl-srw.7
- Yessenbayev, Z., & **Kozhirbayev, Z.** (2024). Use of Riemannian distance metric to verify topological similarity of acoustic and text domains. In *Proceedings of the 33rd International Conference on Artificial Neural Networks (ICANN 2024)* (pp. 368–380). Lugano, Switzerland. Springer. DOI: https://doi.org/10.1007/978-3-031-72350-6_25
- **Kozhirbayev, Z.** (2024). Leveraging Wav2Vec2.0 for Kazakh speech recognition: An experimental study. In *Proceedings of the 24th International Conference on Computational Science and Its Applications, (ICCSA 2024*) (pp. 120–132). Hanoi, Vietnam. Springer. DOI: https://doi.org/10.1007/978-3-031-64608-9_8
- **Kozhirbayev, Z.,** Islamgozhayev, T., Sharipbay, A., Serkazyyeva, A., & Yessenbayev, Z. (2023). Preliminary tasks of word embeddings comparison of unaligned audio and text data for the Kazakh language. In *Proceedings of the IEEE International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT 2023)* (pp. 435–440). Venice, Italy. IEEE. DOI: https://doi.org/10.1109/WI-IAT59888.2023.00072
- Yessenbayev, Z., & **Kozhirbayev, Z.** (2022). Comparison of word embeddings of unaligned audio and text data using persistent homology. In *Proceedings of the International Conference on Speech and Computer (SPECOM 2022)* (pp. 700–711). Gurugram, India. Springer. DOI: https://doi.org/10.1007/978-3-031-20980-2_59
- Yessenbayev, Z., **Kozhirbayev, Z.,** & Makazhanov, A. (2020). KazNLP: A pipeline for automated processing of texts written in Kazakh language. In *Proceedings of the International*

Conference on Speech and Computer (SPECOM 2020) (pp. 657–666). St. Petersburg, Russia. Springer. DOI: https://doi.org/10.1007/978-3-030-60276-5_63

- **Kozhirbayev, Z.,** & Yessenbayev, Z. (2020). Kazakh text normalization using machine translation approaches. In *CEUR Workshop Proceedings* (Vol. 2780, pp. 115-122). CEUR-WS.
- **Kozhirbayev, Z.,** Erol, B. A., Sharipbay, A., & Jamshidi, M. (2018). Speaker recognition for robotic control via an IoT device. In *Proceedings of the World Automation Congress (WAC 2018)* (pp. 1-5). Stevenson, WA, United States. DOI: https://doi.org/10.23919/WAC.2018.8430295